

Vector Autoregression (VAR)

Jakob Sverre Alexandersen
GRA4159 Trends, Cycles & Signals Extraction
Lecture 8

February 25, 2026

Contents

1	Recap	2
2	Why VAR?	2
3	The VAR model	2
3.1	Reduced form	3
3.2	An example	3
3.3	Simulated VAR data	4
3.4	The companion form	4
3.5	Stability of VARs	5
3.6	Moving average representation	6
3.6.1	Moving average by recursive substitution	6
3.6.2	Moving average representation using lag operators	6
3.7	Finding the MA coefficients	7
3.8	The first and second order moments of VAR	7
3.8.1	EV of the VAR	7
3.8.2	Covariance and autocovariance of VAR	8
3.9	The autocorrelation function	8
4	Estimating VARs	8
4.1	Lag selection and testing	9
5	Forecasting with VARs	9
5.1	Point forecasts	9
5.1.1	Convergence of point forecast	9
5.2	Forecast errors and the MSE	10
5.2.1	MSE	10
5.3	The MSE and MA representation	10
5.3.1	Convergence of MSE	10
5.4	Forecast intervals	11
5.5	Forecasting with VARs: an example	11
6	Granger causality	12
6.1	Example	12
7	Summary	13

1 Recap

- We know the difference between stationary and non-stationary variables and time series processes, such as AR, MA, RW
- We also know why we sometimes want to focus on trend and cycle decompositions and how to compute them using different methods

But, we only have been working with univariate models, i.e., just one outcome variable. Surely this is not realistic! Different economic financial variables affect each other, right??

2 Why VAR?

- Multivariate extension of VAR
- Workhorse model for empirical macro economics
- Can model many time series simultaneously within a single model
 - For instance, GDP depends on inflation, and inflation depends on GDP
- Facilitates multivariate forecasts
 - For instance, conditional on knowing GDP and inflation today, what do we expect them to be next quarter?
- Facilitates structural identification. Can do causal inference
- For instance, if GDP increases by 1%, what happens to inflation (taking everything else as given)?
- Will not cover the causal part in this course

3 The VAR model

A VAR(p) model, of order p , can be written as:

$$y_t = \mu + A_1 y_{t-1} + A_2 y_{t-2} + \dots + A_p y_{t-p} + e_t$$

where y_t is a $(K \times 1)$ vector of rvs:

$$y_t = (y_{1,t}, \dots, y_{K,t})'$$

and

$$\mathbb{E}[e_t e_s'] = \begin{cases} \Sigma_e & \text{for } t = s \\ 0 & \text{otherwise} \end{cases} \quad \mathbb{E}[e_t] = 0$$

summarizes the properties of the error terms (i.e., Gaussian white noise)

3.1 Reduced form

- The VAR(p) written as:

$$y_t = \mu + A_1 y_{t-1} + A_2 y_{t-2} + \cdots + A_p y_{t-p} + e_t$$

is called a reduced form representation

- This separates it from the structural VAR (SVAR), which is used for causal inference, and not covered here

3.2 An example

- Assuming $K = 2$ and $p = 1$, we have the VAR(1) model:

$$y_t = \mu + A_1 y_{t-1} + e_t$$

where y_t, μ, e_t and the A_1 matrices are given as:

$$y_t = \begin{bmatrix} y_{1,t} \\ y_{2,t} \end{bmatrix}, \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, A_1 = \begin{bmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \end{bmatrix}, e_t = \begin{bmatrix} e_{1,t} \\ e_{2,t} \end{bmatrix}$$

- As an example, assume that the specific elements of the A_1 matrix are known and given as:

$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0.5 & 0 \\ 1 & 0.2 \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + \begin{bmatrix} e_{1,t} \\ e_{2,t} \end{bmatrix}$$

after doing the matrix multiplications, we can write

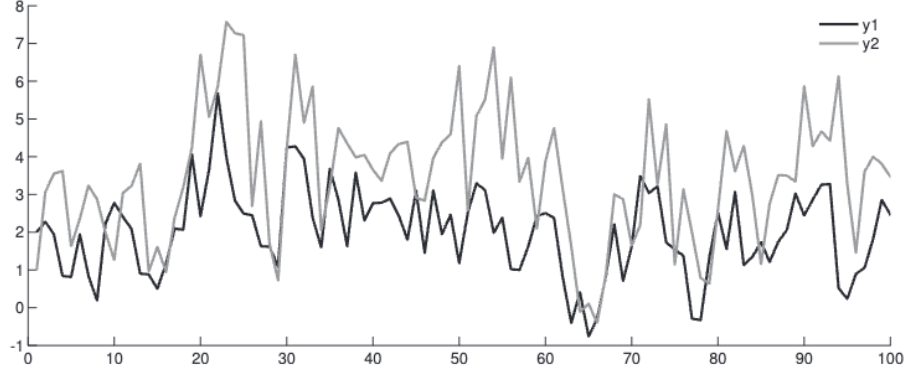
$$\begin{aligned} y_{1,t} &= 1 + 0.5y_{1,t-1} + e_{1,t} \\ y_{2,t} &= 1 + y_{1,t-1} + 0.2y_{2,t-1} + e_{2,t} \end{aligned}$$

- Assume further:

$$\begin{bmatrix} e_{1,t} \\ e_{2,t} \end{bmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0.2 \\ 0.2 & 1 \end{bmatrix} \right)$$

3.3 Simulated VAR data

Figure: Simulated series using a VAR(1)



As the process is stable by definition, more on this later, both series $y_{1,t}$ and $y_{2,t}$ fluctuate around a constant mean, and their variability does not change as they wander along

3.4 The companion form

- VAR(p) models, with $p > 1$, can be written as VAR(1) models by employing the so-called companion form
- Using the companion form simplifies notation and some derivations
- A VAR(p):

$$y_t = \mu + A_1 y_{t-1} + A_2 y_{t-2} + \cdots + A_p y_{t-p} + e_t$$

can be written as a VAR(1):

$$\vec{Z}_t = \Gamma_0 + \Gamma_1 Z_{t-1} + v_t$$

- How? Stack the vector $(y_t, y_{t-1}, \dots, y_{t-p})$ in the $Kp \times 1$ vector $\vec{Z}_t$ as follows:

$$\vec{Z}_t = \begin{bmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{bmatrix} = \begin{bmatrix} y_{1,t} \\ y_{2,t} \\ \vdots \\ y_{K,t} \\ \vdots \\ y_{1,t-p+1} \\ y_{2,t-p+1} \\ \vdots \\ y_{K,t-p+1} \end{bmatrix}$$

- Writing out the matrices:

$$\begin{bmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{bmatrix} = \begin{bmatrix} \mu \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \begin{bmatrix} A_1 & A_2 & \dots & A_{p-1} & A_p \\ I & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & 0 & 0 \\ 0 & 0 & \dots & I & 0 \end{bmatrix} \begin{bmatrix} y_{t-1} \\ y_{t-2} \\ \vdots \\ y_{t-p} \end{bmatrix} + \begin{bmatrix} e_t \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

- Accordingly, the dimensions of the vectors Z_t, Γ_0 and v_t are $Kp \times 1$. A_j for $j = 1, \dots, p$ is $K \times K$, and Γ_1 is $Kp \times Kp$
- Γ_1 is the companion form matrix
- We also covered this when talking about AR processes. Same thing here, but multivariate.

3.5 Stability of VARs

- As in the multivariate case $AR(p)$, we most often want the $VAR(p)$ to be covariance stationary
- For this to happen, the effects of shocks e_t must die out
- How to check for stationarity?

Stationarity:

Stationarity $\iff$ eigenvalues of the companion form matrix are all less than one in absolute value

- The eigenvalues are those numbers λ for which:

$$|\Gamma_1 - \lambda I| = 0$$

- Example, using numeric values from above:

$$\begin{aligned} \det \left(\begin{bmatrix} 0.5 & 0 \\ 1 & 0.2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) &= \det \left(\begin{bmatrix} 0.5 - \lambda & 0 \\ 1 & 0.2 - \lambda \end{bmatrix} \right) \\ &= (0.5 - \lambda)(0.2 - \lambda) = 0 \\ &\rightarrow \lambda_1 = 0.5, \quad \lambda_2 = 0.2 \end{aligned}$$

3.6 Moving average representation

- Assuming that the VAR(p) model is stable, we can derive its infinite moving average representation
- Two ways: recursive substitution or using lag operators
- Again, this mirrors the result for the univariate AR case

3.6.1 Moving average by recursive substitution

- Assume VAR(1):

$$y_t = \mu + A_1 Y_{t-1} + e_t$$

- Recursive substitution gives:

$$y_t = \left(I + A_1 + A_1^2 + \dots + A_1^j \right) \mu + A_1^{j+1} y_{t-(j+1)} + e_t + A_1 e_{t-1} + A_1^2 e_{t-2} + \dots + A_1^j e_{t-j}$$

where $A_1^0 = I$

- Stability implies $\left(I + A_1 + A_1^2 + \dots + A_1^j \right) \mu \rightarrow (I - A_1)^{-1} \mu$ as $j \rightarrow \infty$, and, $A_1^{j+1} y_{t-(j+1)} \rightarrow 0$ as $j \rightarrow \infty$
- Hence,

$$y_t = v + \sum_{j=0}^{\infty} A_1^j e_{t-j}$$

where $v = (I - A_1)^{-1} \mu$

3.6.2 Moving average representation using lag operators

- VAR(1) using lag operators:

$$(I - A_1 L) y_t = \mu + e_t$$

which is often written:

$$A(L) y_t = \mu + e$$

Thus, $(I - A_1 L) = A(L)$ (VAR(p)):

$$A(L) = (I - A_1 L - A_2 L^2 - \dots - A_p L^p)$$

- Multiplying with $A(L)^{-1}$, we get the vector moving average representation:

$$\begin{aligned} y_t &= A(L)^{-1} \mu + A(L)^{-1} e_t = B(L) \mu + B(L) e \\ &= v + \sum_{j=0}^{\infty} B_j e_{t-j} \end{aligned}$$

- using the geometric rule:

$$A(L)^{-1} = (I - A_1 L)^{-1} = \sum_{j=0}^{\infty} A_1^j L^j \equiv B(L) = \sum_{j=0}^{\infty} B_j L^j$$

with $B_0 = I$ and $v = \left(\sum_{j=0}^{\infty} B_j \right) \mu$

3.7 Finding the MA coefficients

- The VAR(1) written in terms of the moving average coefficients:

$$y_t = v + \sum_{j=0}^{\infty} B_j e_{t-j}$$

where $B_j = A_1^j$, and $B_0 = I$

- How can we determine the B_j s, i.e., the moving average coefficients?
- Use the relationship $I = B(L)A(L)$, which holds due to the result above:

$$A(L)^{-1} = B(L) \Rightarrow I = B(L)A(L)$$

because $A(L)^{-1}A(L) = I$. Thus,

$$\begin{aligned} I &= (B_0 + B_1L + B_2L^2 + \dots) (I - A_1L - A_2L^2 - \dots - A_pL^p) \\ &= (B_0 + B_1L + B_2L^2 + \dots) - (B_0A_1L + B_1A_1L^2 + B_2A_1L^3 + \dots) - (B_0A_2L^2 + B_1A_2L^3 + \dots) - \dots \\ &= B_0 + (B_1 - B_0A_1)L + (B_2 - B_1A_1 - B_0A_2)L^2 + \dots + \left(B_i - \sum_{j=1}^{\infty} B_{i-j}A_j \right) L^i + \dots \end{aligned}$$

Solving for the relevant lags (noting that $A_j = 0$ for $j > p$), we get:

$$\begin{aligned} L^0 : B_0 &= I \\ L^1 : B_1 &= B_0A_1 \\ L^2 : B_2 &= B_1A_1 + B_0A_2 \\ &\Downarrow \\ L^i : B_i &= \sum_{j=1}^i B_{i-j}A_j \quad \text{for } i = 1, 2, \dots \end{aligned}$$

3.8 The first and second order moments of VAR

From above: MA representation gives

$$y_t = v + \sum_{j=0}^{\infty} B_j e_{t-j}$$

with $v = \left(\sum_{j=1}^{\infty} B_j \right) \mu = (I - A_1)^{-1} \mu$

3.8.1 EV of the VAR

Due to the Gaussian white noise assumption on the error terms, the EV of the process is simply:

$$\mathbb{E}[y_t] = v = (I - A_1)^{-1} \mu$$

This mean is sometimes described as the steady-state of the system

3.8.2 Covariance and autocovariance of VAR

Define the covariance and autocovariance of the process by Y , which equals:

$$\begin{aligned} Y(s) &= \mathbb{E}[(y_t - v)(y_{t-s} - v)'] \\ &= \mathbb{E} \left[\left(\sum_{j=0}^{s-1} B_j e_{j-t} + \sum_{j=0}^{\infty} B_{s+j} e_{t-s-j} \right) \left(\sum_{j=0}^{\infty} B_j e_{t-s-j} \right)' \right] \\ &= \sum_{j=0}^{\infty} B_{s+j} \Sigma_e B_j' \end{aligned}$$

3.9 The autocorrelation function

Normalization of $Y(s)$ gives the autocorrelation function:

$$R(s) = D^{-1} Y(s) D^{-1}$$

where D is the square root of the diagonal elements of $Y(0)$

4 Estimating VARs

- Can be done equation by equation using OLS: consistent and efficient when the errors are normally distributed
- More formally, define: $Y = [y_1, \dots, y_T]$, $A = [A_1 : \dots : A_p]$, $U = [e_1, \dots, e_T]$ and $Z = [Z_0, \dots, Z_{T-1}]$ where:

$$Z_{t-1} = \begin{bmatrix} y_{t-1} \\ y_{t-2} \\ \vdots \\ y_{t-p} \end{bmatrix}$$

- The model can be written as

$$Y = AZ + U$$

and the OLS estimator of A is defined as:

$$\hat{A} = [\hat{A}_1 : \dots : \hat{A}_p] = Y Z' (Z Z')^{-1}$$

- Under standard assumptions, the OLS estimator $\hat{A}$ is consistent and asymptotically normally distributed:

$$\sqrt{T} \text{vec}(\hat{A} - A) \xrightarrow{d} N(0, \Gamma^{-1} \otimes \Sigma_e)$$

where $\xrightarrow{d}$ implies convergence in distribution and $Z Z' / T \xrightarrow{p} \Gamma$

- Accordingly, standard inference applies to the estimated VAR parameters
- When no parameter restrictions are employed on the VAR system, and for a normally distributed Gaussian stationary process y_t , the OLS estimator is identical to the MLE conditional on the initial values.

4.1 Lag selection and testing

- VARs can be estimated equation by equation using OLS
- VAR models are heavily parameterized:
 - Significant tests on individual coefficients not necessarily very informative (depends on usage)
 - More important to check properties of residuals: does the lag structure correctly capture the dynamics?
- How many lags?
 - Use information criteria (BIC, AIC, etc)
 - Check properties of residuals (i.e., whiteness and autocorrelation)

5 Forecasting with VARs

- Again, this mirrors the results for the univariate AR case
- Iterating the VAR forward:

$$y_{t+h} = (I + A_1 + \dots + A_1^{h-1}) \mu + A_1^h y_t + \sum_{i=0}^{h-1} A_1^i e_{t+h-i}$$

- Thus, the forecast $h = 01, \dots, H$ is a function of $I(t)$ (the information set at time t only)
- Note: the above equation is valid only for a VAR(1) process. Use companion form for more degrees processes and the formula remains unchanged

5.1 Point forecasts

- By employing the conditional expectation to the equation above, we get the VAR point forecast:

$$\mathbb{E}[y_{t+h}|y_t] = (I + A_1 + \dots + A_1^{h-1}) \mu + A_1^h y_t$$

- $\mathbb{E}[y_{t+h}|y_t]$ is often called the predictor and is simply denoted $y(h)$

5.1.1 Convergence of point forecast

Under the maintained assumption about stationarity, the point forecasts converge to the unconditional mean of the process when h goes to infinity:

$$\mathbb{E}[y_{t+h}|y_t] = \frac{\mu}{I - A_1} \text{ when } h \rightarrow \infty$$

Again, these equations are just the multivariate extensions of the univariate case

5.2 Forecast errors and the MSE

- We are also interested in the forecast error and its variance, i.e., the MSE
- The VAR forecast error at horizon h is:

$$y_{t+h} - y(h) = \sum_{i=0}^{h-1} A_1^i e_{t+h-i}$$

Using the expectations operator:

$$\mathbb{E}[y_{t+h} - y(h)] = \mathbb{E}[y_{t+h}] - \mathbb{E}[y(h)] = 0$$

Thus, the predictor $y(h)$ is unbiased

5.2.1 MSE

- The MSE is simply the forecast error variance, denoted $\Sigma_y(h)$:

$$\Sigma_y(h) = \mathbb{E}[(y_{t+h} - y(h))^2] = \mathbb{E}\left[\left(\sum_{i=0}^{h-1} A_1^i e_{t+h-i}\right)\left(\sum_{i=0}^{h-1} A_1^i e_{t+h-i}\right)'\right]$$

$\mathbb{E}[e_{t+h-i}] = \Sigma_e$ for all h , thus:

$$\Sigma_y(h) = \sum_{i=0}^{h-1} (A_1^i \Sigma_e A_1^{i'})$$

5.3 The MSE and MA representation

- Expanding

$$\begin{aligned} y_t &= A(L)^{-1}\mu + A(L)^{-1}e_t = B(L)\mu + B(L)e \\ &= v \sum_{j=0}^{\infty} B_j e_{t-j} \end{aligned}$$

and move it h periods forwards, we get:

$$y_{t+h} = v + e_{t+h} + B_1 e_{t+h-1} + B_2 e_{t+h-2} + \dots + B_{h-1} e_{t+1} + B_h e_t + B_{h+1} e_{t-1} + \dots$$

Computing the forecast gives:

$$y_{t+h} - \mathbb{E}[y_{t+h} - \mathbb{E}[y_{t+h}|y_t]](y_{t+h} - \mathbb{E}[y_{t+h}|y_t])' = \sum_{j=0}^{h-1} (B_j \Sigma_e B_j')$$

5.3.1 Convergence of MSE

When $h \rightarrow \infty$, the equation above is equal to $Y(0)$, the unconditional variance of the VAR process. In mathematical terms:

$$\Sigma_y(h) = Y(0) \text{ when } h \rightarrow \infty$$

5.4 Forecast intervals

- Assuming that the errors of the VAR model are Gaussian, i.e., $e_t \sim N(0, \Sigma_e)$, and independent across time, implies that the forecast errors are also normally distributed, such that:

$$\frac{y_{k,t+h} - y(k, h)}{\sigma_{y,k}(h)} \sim N(0, 1)$$

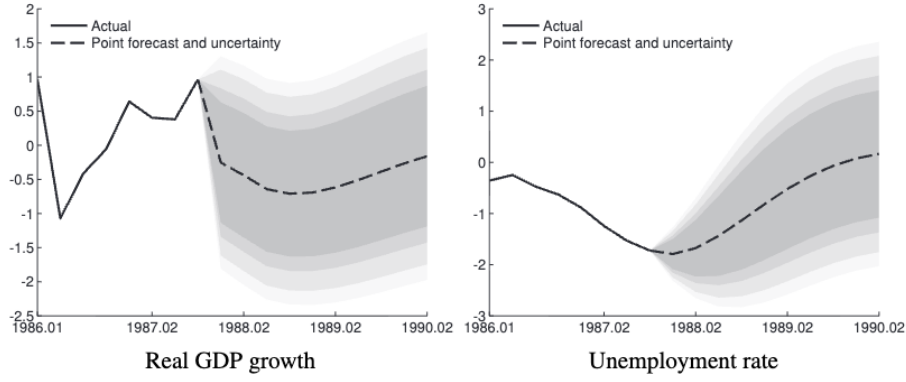
where $y_{k,t+h}$ and $y(k, h)$ are the k 'th elements of the actual and predictor values, respectively, and $\sigma_{y,k}(h)$ is the square root of the variance for the k 'th equation. That is, the square root of the k 'th element on the diagonal of the $\Sigma_y(h)$ matrix.

- Forecast intervals can be generated around the VAR point forecasts using

$$[y(k, h) - z_{\alpha/2}\sigma_{y,k}(h), y(k, h) + z_{\alpha/2}\sigma_{y,k}(h)]$$

5.5 Forecasting with VARs: an example

Figure: Forecasting a VAR: GDP growth and unemployment



Note: The fan charts correspond to the 30, 50, 70 and 90 percent quantiles of the forecast distribution, with the darkest shade corresponding to the smallest quantile (30 percent).

6 Granger causality

- Cause cannot come after effect $\Rightarrow$ if x causes y , x should help predict y
- Granger causality $\Rightarrow$ predictability test!
- Consider the VAR system:

$$A_1 = \begin{bmatrix} 0.5 & 0 \\ 1 & 0.2 \end{bmatrix}$$

$y_{1,t}$ is not a function of past $y_{2,t}$, i.e., $y_{2,t}$ does not help predict $y_{1,t}$. However, $y_{2,t}$ is a function of past $y_{1,t}$, and therefore $y_{1,t}$ does help predict $y_{2,t}$. Accordingly, following the definition of Granger causality, $y_{1,t}$ does Granger cause $y_{2,t}$, but $y_{2,t}$ does not cause $y_{1,t}$

- Alternatively, if the coefficient matrix had been:

$$A_1 = \begin{bmatrix} 0.5 & 0.2 \\ 1 & 0.2 \end{bmatrix}$$

both $y_{1,t}$ and $y_{2,t}$ would have Granger caused each other. In this case we often say that the system for y_t is a feedback system.

- In practical situations the VAR representation has to be estimated
- A test for Granger causality is implemented by employing a joint hypothesis test:
 - For each equation in the VAR we compare $K - 1$ restricted versions of the VAR model against the unrestricted version
 - The test, for each restricted version, is a joint hypothesis test that all the lags of the k 'th variable in the system are jointly significantly different from zero. As such, this is simply a standard F -test, where the null hypothesis is no Granger causality.

6.1 Example

- Assume a VAR(2) model, with three variables, the first equation of the system reads:

$$y_{1,t} = \mu_1 + \alpha_{11}y_{1,t-1} + \alpha_{12}y_{2,t-1} + \alpha_{13}y_{3,t-1} + \alpha_{14}y_{1,t-2} + \alpha_{15}y_{2,t-2} + \alpha_{16}y_{3,t-2} + e_{1,t}$$

- The test for no Granger causality from $y_{2,t}$ to $y_{1,t}$ would be an F-test with $H_0 : \alpha_{12} = \alpha_{15} = 0$, and the test for no Granger causality from $y_{3,t}$ to $y_{1,t}$ would be an F-test with $H_0 : \alpha_{13} = \alpha_{16} = 0$. Conducting similar tests on the two other equations of the VAR representation would give a complete test for no Granger causality
- A rejection of any of the null hypotheses indicates Granger causality

Specification	Equation	Variable	
		GDP	Unemployment
VAR(2)	GDP	0.75	0.00
	Unemployment	0.00	0.00
VAR(8)	GDP	0.38	0.00
	Unemployment	0.00	0.00

Note: Null hypothesis of no Granger causality. Numbers are p-values. See text for details.

Reject H_0 of no Granger causality for all but in one case: GDP is not Granger causing itself

7 Summary

1. The vector autoregression (VAR) is a multivariate version of the univariate autoregressive model (AR)
2. A $\text{VAR}(p)$ can be written as a first order $\text{VAR}(1)$ model by writing it in the companion form.
3. If all eigenvalues of the companion form matrix are less than 1 in absolute value, the VAR model is stable. A stable VAR model can be inverted and written as an infinite vector moving average model.
4. The VAR can be estimated by OLS equation-by-equation. Under standard assumptions, the OLS estimator will be similar to the MLE of the whole system
5. Testing for Granger causality involves testing whether or not lagged values of a given variable in the VAR system help predict one of the endogenous variables in the system. Such a test can simply be conducted using an F-test, where the null hypothesis is no Granger causality.